

STeP Analytical Performance Model

Each node n in the DAG is characterized by four quantities derived from its operator type and stream shapes:

Symbol	Name	Description
$T_{\text{fire}}(n)$	Firing time	Processing time per input step (roofline)
$N_{\text{fire}}(n)$	Firing count	Total number of firings
$NIT_n^{n'}$	Input tile count	Input tiles consumed from predecessor n' per output chunk
$OTPC(n)$	Output tiles per chunk	Output tiles produced per output chunk

Quantity	Equation
Firing Time (roofline per step)	$T_{\text{fire}}(n) = \max\left(T_{\text{PMU-read}}, \left\lceil \frac{FLOPs}{BW_c} \right\rceil, T_{\text{PMU-store}}\right)$
Start Time (predecessor gating)	$st(n) = \max_{n' \in \text{pred}(n)} (fto(n'))$
First Tile Out (latency to first output)	$fto(n) = st(n) + \underbrace{(NIT - 1) \cdot \max(T_{\text{fire}}, OTI(n'))}_{\text{input chunk delay}} + T_{\text{fire}}(n)$
Output Chunk Interval (steady-state rate)	$OCI(n) = \max_{n' \in \text{pred}(n)} \left(NIT_n^{n'} \cdot \max(T_{\text{fire}}(n), OTI(n')) \right)$
Output Tile Interval	$OTI(n) = \frac{OCI(n)}{OTPC(n)}$
End Time (node completion)	$end(n) = fto(n) + (N_{\text{fire}}(n) - 1) \cdot OTI(n)$
Graph Time (program total)	$T_{\text{graph}} = \max_{n \in \text{leaf}(G)} (end(n))$

Also modeled:

- **HBM contention** — two-pass estimation: solve without contention to find overlapping off-chip ops, then recompute with rate-weighted concurrent requests across shared channels.
- **One-shot predecessor correction** — when a predecessor produces only one output chunk ($N_{\text{fire}}(n') \leq NIT$), its OTI is replaced by T_{fire} to avoid propagating a decoupled rate.
- **Burst output (OTPC > 1)** — expansion and triggered-load operators produce multiple tiles per chunk; OTI distributes the chunk interval across all output tiles.
- **Fan-out / fan-in routing** — FlatPartition multiplies effective OTI by consumer count N_c ; FlatReassemble divides by data input count N_d .
- **Dynamic dimensions** — symbolic variables resolved via expected-value substitution assuming uniform routing, or by concrete trace instantiation.

Validation: Predicted vs. Simulated Cycle Counts (162 configurations)

Kernel	N	Mean	Max	Kernel	N	Mean	Max
outer_product_accum	5	0.9%	2.0%	residual_add_norm	5	2.7%	7.3%
bufferize_chain	5	7.9%	9.9%	qkv_projection	8	0.3%	0.9%
chained_unary	5	1.9%	5.0%	rms_norm	9	1.1%	3.3%
copy_2d	9	0.9%	3.1%	scaled_dot_product	5	2.1%	7.0%
element_wise_add	9	0.3%	1.4%	attention	9	1.9%	9.5%
Mixtral8x7B	1	13.0%	13.0%	moe_expert_single	7	1.2%	4.0%
gated_mlp	7	0.3%	1.7%	moe_routed	21	15.4%	68.2%
gemm	11	0.1%	0.4%	silu_activation	9	0.5%	2.6%
layernorm	5	9.2%	25.8%	softmax	5	2.0%	3.4%
matmul_binarymap	9	0.1%	0.4%	vector_reduce_sum	6	0.7%	2.0%
Overall (162 configs):				Mean error = 3.5%, Median = 0.6%			